JESSE CHEN

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EDUCATION

University of Southern California - GPA 3.81 **May 2025**
B.S. Computer Science

Related Coursework:
Computer Architecture · Cross-Platform App Development · Professional C++ · Intro to OS · Intro to Internetworking · iOS App Development · Software Engineering · Intro to AI · Full-Stack Web Development Algorithms · Intro to Computer Systems · Intro to Embedded Systems Software Development · Data Structures and OOD Mathematics of Machine Learning · Statistics · Discrete Methods in CS Probability Theory · Programming with Data Structures · C++ Programming · Differential Equations · Discrete Structures ·Linear Algebra · Multivariable Calculus · Python

SKILLS

Languages: C++, HTML/CSS, Python, JavaScript, C, C#, Swift, SQL Frameworks: Bootstrap

WORK EXPERIENCE

Software Engineering Intern, AMD (San Jose, CA) **May 2024 - Aug 2024**
▪ Worked with the AI team to help automate and track tests for hardware and monitor relevant metrics and statistics
▪ Helped set-up databases and designed the backend schemas using Python, Shell script and SQL
▪ Populated a QOR Dashboard with graph and table visualizations using Grafana

Teaching Assistant for CS170: Discrete Methods in CS, USC (Los Angeles, CA) **Aug 2023 - May 2024**
▪ Worked with the course Professor to create homework, quizzes, and an overall semester plan for the fall semester
▪ Held office hours and worked with students to help them understand concepts and class material
▪ Worked alongside other course producers and staff to ensure efficiency and team-building

Software Engineering Intern, Uplift Labs, Inc. (Palo Alto, CA) - Swift/Xcode **Jun 2022 - Aug 2022**
▪ Worked with a mentor and other interns to implement sports movement detection using computer vision and AI
▪ Programmed in Swift, using Swift's Vision framework, to develop an iOS app that performed pose estimation and keypoint visualization, including segment drawing and bounding box tracking
▪ Debugged and tested application on iPhone 13 and optimized GUI operations for live video

PROJECTS

CS401 Capstone Project - Interactive Furniture Layout Tool - JavaScript/React **Sep 2024 - Nov 2024**
▪ Developed a React-based interactive room layout visualization tool allowing users to customize furniture arrangements for multi-floor buildings with drag-and-drop functionality, layout presets, dynamic inventory management
▪ Implemented modular state management for floor-specific configurations, supporting unique furniture types and layouts
▪ Enhanced user experience with design choices such as multi-item mouse selection and keyboard shortcuts

Multithreaded Network of Servers - C/C+ **September 2023**
▪ Wrote a multithreaded network of nodes acting as web servers and clients to understand the flow of network communication
▪ Implemented persistent TCP connection with a link layer and a network layer through an overlay network, and added a transport layer with a rudimentary network application to simulate using transport layer services

FindMyClassmates - Java/Android Studio **December 2023**
▪ Developed an application to simulate course registration for students, complete with adding/dropping classes + reviews
▪ Functionalities include login/signup, chatting capabilities between students, and a user profile section complete with editing

Cache Lab - C **March 2023**
▪ Wrote a 300~ line C program that simulates the behavior of a cache memory
▪ Handles a memory trace as input with the implementation of both FIFO and LRU eviction policies

First Place at UC Riverside CutieHack Hackathon - C#/Unity **Nov 2021**
▪ Collaborated with teammates to design, code, and present a functional Unity game within the 12-hour hackathon
▪ Project Page: devpost.com/software/space-slimes

Socially Assistive Robot w/ Raspberry Pi - Python & C#/Unity **Jun 2020 - Aug 2020**
▪ Designed and built a socially assistive robot with a Raspberry Pi and PiCamera, which included functionality such as spatial localization in a 2D plane, scanning QR codes for dynamic commands, and the display of information through AR
▪ Gave a formal presentation and a demo, project website: padlet.com/USCViterbiSTEM/SHINE20_JesseC